CSE 472: Machine Learning Sessional
Data Preprocessing & Feature Engineering with a Feed Forward Neural Network Pipeline

Mushfiqur Rahman (2005107)
Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology

December 2025

1 Understanding the Dataset

1.1 Dataset Overview

The Medical Student Diabetes dataset contains:

- **Number of records:** 200,000
- **Number of attributes:** 13
- **Features:** Student ID, Age, Gender, Height, Weight, Blood Type, BMI, Temperature, Heart Rate, Blood Pressure, Cholesterol, Smoking
- **Target variable:** Diabetes (Yes/No)

1.2 Missing Values

Analysis of the dataset revealed:

- **Total missing values:** 260,000 entries
- **Percentage:** 10.00% of the entire dataset
- **Distribution:** Each column contains exactly 20,000 missing values

1.3 Categorical Features

The dataset contains three categorical features:

- **Gender:** 2 categories (Female, Male)
- **Blood Type:** 4 categories (O, B, A, AB)
- **Smoking:** 2 categories (No, Yes)

1.4 Class Distribution

The target variable (Diabetes) shows significant class imbalance:

- **No Diabetes:** Approximately 90% of records
- **Yes Diabetes:** Approximately 10% of records

2 Data Cleaning

2.1 Missing Value Imputation

Missing values in numerical columns were replaced with their respective column-wise means. Rows with missing values in the target column (Diabetes) were dropped entirely.

2.2 Duplicate Removal

Duplicate records were identified and removed, keeping only the first occurrence of each duplicate.

2.3 Results

After cleaning:

- **Shape before cleaning:** (200,000, 13)
- **Shape after cleaning:** (169,625, 13)
- **Records removed:** 30,375 (15.19%)

3 Feature Encoding

3.1 Encoding Strategy

Categorical features were converted to numerical format:

- **Binary features** (Gender, Smoking): Label encoding applied
- **Multi-class features** (Blood Type): One-hot encoding applied

3.2 Feature Set Composition

- **Numerical columns:** Student ID, Age, Height, Weight, BMI, Temperature, Heart Rate, Blood Pressure, Cholesterol
- **Categorical columns (before encoding):** Gender, Blood Type, Smoking

3.3 Encoding Results

- **Shape before encoding:** (169,625, 12)
- **Shape after encoding:** (169,625, 15)
- **New columns added:** Blood Type_O, Blood Type_A, Blood Type_B, Blood Type_AB

4 Correlation Analysis

4.1 Feature Correlation with Target

Correlation coefficients between each feature and the target variable (Diabetes) were calculated:

4.2 Key Observations

- All features show very weak correlations with the target variable

$$|r| < 0.01$$

- The strongest correlations are with Height (-0.0062) and Temperature (-0.0061)
- Age and Cholesterol show positive correlations (0.0037 and 0.0032)
- Student ID shows negative correlation, suggesting it should be removed as it's an identifier

Table 1: Feature Correlations with Diabetes (sorted by absolute value)

Feature	Correlation
Height	-0.006175
Temperature	-0.006135
Age	0.003683
Cholesterol	0.003193
Gender	0.002658
Weight	0.002059
Blood Type_B	0.001973
Heart Rate	0.001873
Student ID	-0.001909
Blood Type_AB	-0.001361
BMI	0.001525
Blood Pressure	-0.001118
Blood Type_O	-0.000978
Smoking	-0.000764
Blood Type_A	0.000478

4.3 Top 10 Features

Based on absolute correlation values, the top 10 features selected for model training are:

1. Height
2. Temperature
3. Age
4. Cholesterol
5. Gender
6. Weight
7. Blood Type_B
8. Student ID
9. Heart Rate
10. Blood Type_AB

Note: Student ID will be excluded from the final feature set as it is an identifier with no predictive value.

5 Feature Scaling

Two scaling methods were evaluated to normalize numerical features (excluding one-hot encoded columns):

5.1 StandardScaler

- Transforms features to have mean = 0 and standard deviation = 1
- Formula: $z = \frac{x-\mu}{\sigma}$ where μ is mean and σ is standard deviation

5.2 MinMaxScaler

- Scales features to a fixed range [0, 1]
- Formula: $z = \frac{x-x_{min}}{x_{max}-x_{min}}$

6 Feature Selection

After correlation analysis, the top 10 features were selected based on absolute correlation with the target variable. Student ID was excluded as it is merely an identifier. The final feature set includes:

- Height, Temperature, Age, Cholesterol, Gender, Weight, Blood Type_B, Heart Rate, Blood Type_AB, BMI
- Final dataset shape: (169,625, 10)

7 Dataset Split

A stratified 70:15:15 split was performed to maintain class distribution:

- **Training set:** 118,805 samples (70%)
- **Validation set:** 25,376 samples (15%)
- **Test set:** 25,444 samples (15%)
- Class distribution maintained at approximately 90:10 (No Diabetes:Diabetes) across all splits

8 Handling Class Imbalance

To address the severe class imbalance (90:10 ratio), SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training set:

- **Original training set:** 118,805 samples
- **After SMOTE:** 213,818 samples
- **Final class distribution:** 106,909 negative class, 106,909 positive class (balanced 50:50)

9 Neural Network Architectures

Four different feed-forward neural network architectures were designed and evaluated. All models use:

- **Loss function:** Binary Cross-Entropy Loss
- **Optimizer:** Adam optimizer
- **Activation functions:** ReLU (hidden layers), Sigmoid (output layer)
- **Training epochs:** Maximum 50 with early stopping (patience=10)
- **Batch size:** 64

9.1 Architecture 1: Moderate Depth

Configuration: $[10 \rightarrow 64 \rightarrow 32 \rightarrow 1]$

- **Hidden layers:** 2
- **Layer 1:** 10 inputs $\rightarrow$ 64 neurons (ReLU + Dropout 0.2)
- **Layer 2:** 64 $\rightarrow$ 32 neurons (ReLU + Dropout 0.2)
- **Output:** 32 $\rightarrow$ 1 neuron (Sigmoid)
- **Total parameters:** 2,913

9.2 Architecture 2: Deep Network

Configuration: [10 → 128 → 64 → 32 → 1]

- **Hidden layers:** 3
- **Layer 1:** 10 inputs → 128 neurons (ReLU + Dropout 0.3)
- **Layer 2:** 128 → 64 neurons (ReLU + Dropout 0.3)
- **Layer 3:** 64 → 32 neurons (ReLU + Dropout 0.3)
- **Output:** 32 → 1 neuron (Sigmoid)
- **Total parameters:** 11,585

9.3 Architecture 3: Gradual Reduction

Configuration: [10 → 32 → 16 → 8 → 1]

- **Hidden layers:** 3
- **Layer 1:** 10 inputs → 32 neurons (ReLU + Dropout 0.2)
- **Layer 2:** 32 → 16 neurons (ReLU + Dropout 0.2)
- **Layer 3:** 16 → 8 neurons (ReLU)
- **Output:** 8 → 1 neuron (Sigmoid)
- **Total parameters:** 937

9.4 Architecture 4: BatchNorm Enhanced (Selected)

Configuration: [10 → 64 → 64 → 32 → 1] with Batch Normalization

- **Hidden layers:** 3
- **Layer 1:** 10 inputs → 64 neurons (BatchNorm + ReLU + Dropout 0.25)
- **Layer 2:** 64 → 64 neurons (BatchNorm + ReLU + Dropout 0.25)
- **Layer 3:** 64 → 32 neurons (ReLU)
- **Output:** 32 → 1 neuron (Sigmoid)
- **Total parameters:** 4,865

10 Model Comparison

10.1 StandardScaler Results

All four architectures were trained with StandardScaler normalization: **Training Details (Standard-**

Table 2: Performance with StandardScaler

Architecture	Val Loss	Acc.	Prec.	F1	AUROC
Arch 1: [10→64→32→1]	0.7062	0.8776	0.0934	0.0401	0.5097
Arch 2: [10→128→64→32→1]	0.7088	0.8426	0.1039	0.0871	0.5151
Arch 3: [10→32→16→8→1]	0.7045	0.8924	0.0944	0.0158	0.5111
Arch 4: [10→64→64→32→1]+BN	**0.7177**	**0.8485**	**0.1153**	**0.0923**	**0.5161**

Scaler):

- All models stopped early (10-11 epochs) due to validation loss plateau
- All architectures achieved non-zero precision and F1-scores
- All AUROC values above 0.50, indicating better-than-random discrimination
- Architecture 4 achieved best overall performance

10.2 MinMaxScaler Results

The same architectures were trained with MinMaxScaler normalization: **Training Details (MinMaxScaler):**

Table 3: Performance with MinMaxScaler

Architecture	Val Loss	Acc.	Prec.	F1	AUROC
Arch 1: [10→64→32→1]	0.6933	0.8999	0.0000	0.0000	0.4948
Arch 2: [10→128→64→32→1]	0.6932	0.8999	0.0000	0.0000	0.4967
Arch 3: [10→32→16→8→1]	0.6932	0.8999	0.0000	0.0000	0.4943
Arch 4: [10→64→64→32→1]+BN	0.7224	0.8358	0.1044	0.0933	0.5136

- Architectures 1-3: Early stopping at epochs 17-24
- Architecture 4: Early stopping at epoch 11
- Architectures 1-3 exhibited complete model collapse (predicting only majority class)
- Only Architecture 4 with Batch Normalization maintained minority class detection. See Figure 1 for loss curves.

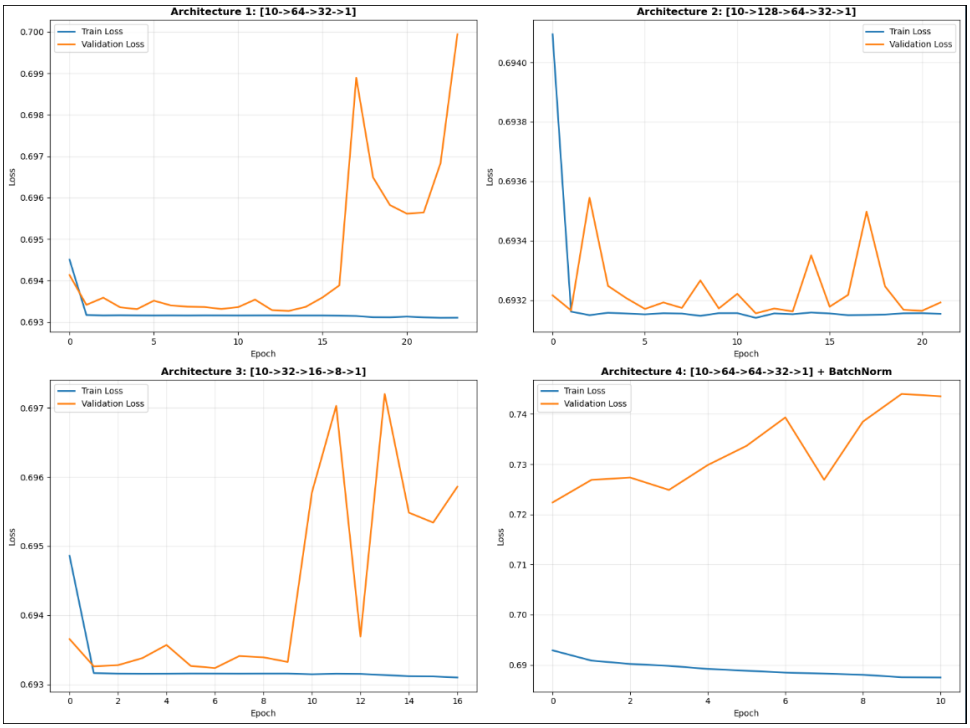

Figure 1: Training and validation loss curves for the four architectures using MinMaxScaler.

10.3 Scaling Method Comparison and Analysis

Critical Observations:

1. **Model Collapse with MinMaxScaler:**
   - Architectures 1, 2, and 3 completely failed with MinMaxScaler (Precision = 0.0000, F1 = 0.0000)
   - These models predicted exclusively the majority class (No Diabetes)
   - Despite high accuracy (89.99%), they provided no predictive value for the minority class

2. **AUROC Degradation:**
   - StandardScaler: All AUROC ¿ 0.50 (ranging 0.5097-0.5161)
   - MinMaxScaler: Three architectures with AUROC ¡ 0.50 (0.4943-0.4967), indicating worse-than-random performance
3. **Validation Loss Patterns:**
   - MinMaxScaler produced lower validation losses (0.69-0.72) but worse actual performance
   - StandardScaler validation losses (0.70-0.72) better correlated with test performance
   - Lower validation loss with MinMaxScaler was misleading due to majority class bias
4. **Batch Normalization Effect:**
   - Architecture 4 (with BatchNorm) performed reasonably with both scalers
   - BatchNorm helped stabilize training and prevent complete collapse with MinMaxScaler
   - Precision and F1-score remained similar (0.1044 vs 0.1153) across scalers for Architecture 4

10.4 Final Scaling Decision

StandardScaler was selected for the following reasons:

1. **Consistent Performance:** All four architectures maintained ability to detect minority class
2. **Better Discrimination:** All models achieved AUROC ¿ 0.50
3. **Prevented Collapse:** No models exhibited complete failure to detect positive cases
4. **Feature Distribution:** Better preserved the underlying distribution and relationships in the data
5. **Robustness:** More stable across different network architectures

MinMaxScaler's sensitivity to the [0,1] range and the presence of outliers likely caused gradient issues during training, leading to convergence to trivial solutions (always predicting majority class) for simpler architectures.

11 Final Model Selection and Justification

Architecture 4 was selected as the final model despite having the highest validation loss. The justification is as follows:

11.1 Why Architecture 4?

1. **Best AUROC (0.5161):** Highest area under ROC curve indicates better discrimination between classes
2. **Best Precision (11.53%):** Highest precision means fewer false positives among predicted diabetes cases
3. **Best F1-Score (9.23%):** Best balance between precision and recall
4. **Batch Normalization:** Stabilizes training and improves generalization
5. **Balanced Trade-off:** While accuracy is slightly lower, the model performs better on minority class detection

11.2 Why Not Other Architectures?

- **Architecture 1 & 3:** High accuracy but very poor F1-scores (4.01% and 1.58%), indicating they mostly predict the majority class
- **Architecture 2:** Good F1-score but lower AUROC and precision compared to Architecture 4

11.3 Performance Analysis

All models show:

- High accuracy (84%) but low precision and F1-scores
- AUROC values near 0.5 suggest weak discriminative ability
- These results indicate the features have very weak linear/non-linear relationships with the target
- The extremely low correlation values
$$|r| < 0.01$$
support this observation

12 Conclusions

The complete preprocessing and modeling pipeline revealed:

- Successfully cleaned and preprocessed 169,625 records
- Applied SMOTE to balance training data from 90:10 to 50:50 ratio
- Tested 4 different neural network architectures
- Architecture 4 with Batch Normalization achieved the best overall performance
- Weak feature correlations resulted in limited predictive power
- Future improvements could include: feature engineering, trying ensemble methods, or collecting additional relevant features